

Irrationality of $F(31/4)$ and the Exact Normalized Hankel Order

Zudilin's forms at rational bases and a unique least-order moment configuration

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ABSTRACT

Specialising Zudilin's 2004 construction for q -harmonic series to rational bases, we prove that for coprime integers $a > b \geq 1$ the Lambert value $F(a/b) = \sum_{n \geq 1} ((a/b)^n - 1)^{-1}$ is irrational whenever $\log b / \log a < \theta^* = 0.4056830213840605 \dots$, the reciprocal of Zudilin's integer-base exponent bound $2.46497868 \dots$. In particular, $F((31/4)^r)$ is irrational for every integer $r \geq 1$, and the same forms bound its irrationality exponent uniformly in r . The linear forms, their permutation-group integrality, the direction and the constants come from Zudilin's 2004 paper. The rational specialisation is this paper's own work: it uses the polynomial form of Lemma 7 there before the integer specialisation, cancels cyclotomic factors before clearing the rational-base denominator, and proves the degree and remainder estimates that measure the cost. The proof is an ordinary proof, and Lean checks the theorem and the corollary, with the source supply applied. The argument does not reach $3/2$. Zudilin bounded the q -order of his 2016 normalised Hankel determinant V_N^* from below; we prove

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6}, \quad [q^{\text{ord}_q V_N^*}] V_N^* = \frac{(N!)^2 (N+1)!}{2^N}.$$

The exponent and coefficient come from a unique least-order tuple in a formal moment expansion. A quantitative selector criterion separates local divisibility from the additional estimates needed for a small nonzero real remainder.

1 Introduction

For $t > 1$, expansion of each geometric series gives

$$F(t) = \sum_{n \geq 1} \frac{1}{t^n - 1} = \sum_{n \geq 1} \frac{\tau(n)}{t^n},$$

where $\tau(n)$ is the number of positive divisors of n . Chowla conjectured that $F(t)$ is irrational for every rational $t > 1$, as Erdős records [1, p. 102]; this is Erdős Problem #1049.

* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Specialising Zudilin's 2004 construction to rational bases, Theorem 2.1 proves irrationality on an explicit region that contains the power family in the title.

Cyclotomic cancellation in the linear forms of Zudilin's 2004 construction [4] produces positive forms $\Lambda_n = U_n F - V_n$ with integral coefficient polynomials of degree at most W_n , and

$$\log(b^{W_n} \Lambda_n(a/b)) = (C_1 \log b - C_0 \log a) n^2 + o(n^2).$$

A negative exponent yields positive integral linear forms tending to zero.

The order of these operations explains the region. At a rational base a/b , each remaining polynomial degree costs a power of b when denominators are cleared. Cancelling a common cyclotomic factor first reduces that degree. The proof must then compare the actual cancelled degree with the decay of the positive remainder; divisibility alone gives neither estimate.

The Hankel argument concerns a different family. Its moment expansion makes one increasing index tuple responsible for the first coefficient. The final sections explain what local cancellation can establish at $3/2$ and specify the real estimate that would complete that approach. The complete catalogue of other constructions and their failure witnesses is retained in the long record.

2 A rational base at which F is irrational

The direction $(14, 12, 14; 27)$, the thirteen intervals and the constants C_1, C_0 below are those of Zudilin's construction [4, §5, pp. 161–162]; there $C_1/C_0 = 2.46497868 \dots$ is his bound for the irrationality exponent at integer bases [4, Thm. 1, p. 154]. Write $\psi_1(u) = \sum_{k \geq 0} (k+u)^{-2}$. Let $\mathcal{J}$ be the thirteen intervals listed in the proof and put

$$C_1 = \frac{1091}{2}, \quad J = \sum_{[u,v] \in \mathcal{J}} (\psi_1(u) - \psi_1(v)), \quad C_0 = 266 - \frac{3}{\pi^2} (225 - J).$$

Thus $\theta^* = C_0/C_1$ and $\mu = C_1/C_0$ are defined exactly.

Theorem 2.1 (rational-base region for Zudilin's forms). *Let $a > b \geq 1$ be coprime integers with*

$$b^\mu < a, \quad \text{equivalently} \quad \frac{\log b}{\log a} < \theta^*,$$

$$\theta^* = \frac{C_0}{C_1} = 0.4056830213840605 \dots,$$

$$\mu = \frac{C_1}{C_0} = 2.4649786835749750 \dots$$

Then $F(a/b)$ is irrational.

The cancellation is performed before homogenisation, so the clearing degree is the degree of the cancelled pair. The value $3/2$ is outside this sufficient region: $J \leq \psi_1(1/14) - \psi_1(1) < 196$ gives $\theta^* < 266/(1091/2) < 1/2 < \log 2 / \log 3$.

Proof of Theorem 2.1. Set

$$a_0 = 14n + 1, \quad a_1 = 12n + 1, \quad a_2 = 14n + 1, \quad \beta = 27n + 2, \quad N = 15n.$$

Use the coefficient pair A_n, B_n of the source identities [4, (8)–(11), pp. 156–157], with $H_n = A_n F - B_n$. Put $D_N(X) = \prod_{\ell=1}^N \Phi_\ell(X)$, $M_n = 266n^2 + 34n + 1$, and

$$\Omega_n(X) = \prod_{\ell=2}^N \Phi_\ell(X)^{\nu_\ell}, \quad \nu_\ell = \omega(n/\ell),$$

where the periodic function

$$\omega(x) = \max\{0, \lfloor 14x \rfloor + \lfloor 13x \rfloor - \lfloor 12x \rfloor - \lfloor 15x \rfloor, \\ 2\lfloor 14x \rfloor - \lfloor 13x \rfloor - \lfloor 15x \rfloor\}$$

is zero or one. Its support in $[0, 1)$ consists of

$$\begin{aligned} &[1/14, 1/12), [1/7, 1/6), [3/14, 1/4), [2/7, 1/3), \\ &[5/14, 2/5), [3/7, 7/15), [1/2, 8/15), [4/7, 3/5), \\ &[9/14, 2/3), [5/7, 11/15), [11/14, 4/5), \\ &[6/7, 13/15), [13/14, 14/15). \end{aligned}$$

These are the intervals $\mathcal{I}$ used to define J . The function ω , these thirteen intervals and the exponents $\nu_\ell = \omega(n/\ell)$ of the source's (22) are printed at [4, pp. 161–162]. The zero-one values and this thirteen-interval support are [omega indicator](#).

Integral polynomials and their degrees. The source's polynomial inclusion is Lemma 7, display (23) [4, p. 161]. Its parameter vector is $n(13, 14, 12, 14, 15, 13)$, its maximum is $15n$, and $\beta - a_1 - a_2 = n > 0$, which is the positivity condition $s > 0$ of the lemma. The conditions (14) of [4, p. 157], $a_1 \leq a_2$ and $a_1 + a_2 \leq \beta \leq a_0 + a_2$, hold as well, so the exponent in (23) is the integer (16), which equals M_n . The lemma's hypotheses therefore hold for every $n \geq 1$. It gives

$$\Lambda_n(X) = X^{-M_n} \frac{D_N(X)}{\Omega_n(X)} H_n(X) = U_n(X) F(X) - V_n(X), \quad U_n, V_n \in \mathbb{Z}[X].$$

This is the polynomial conclusion before the source's subsequent integer-specialisation step in display (24) on p. 162. The coefficientwise interpretation also follows by comparing the source rational coefficients: F is not a rational function, since $F(e^h) = h^{-1} \log(1/h) + O(h^{-1})$ as $h \downarrow 0$. For this estimate split the original sum at $n = 1/h$; the remaining geometric tail is $O(h^{-1})$.

In the source expression (10) for A_n , with coefficients (8) on p. 156, the degree of the k th summand is

$$d_k = a_0 k + E_k + (a_1 - 1)(k - a_1) + (\beta - k - 1)(k - a_2),$$

where

$$E_k = \frac{a_1(a_1 - 1) - (\beta - a_2)(\beta - a_2 - 1) + (\beta - k)(\beta - k - 1)}{2}.$$

For $a_2 \leq k \leq \beta - 2$, one has $d_{k+1} - d_k = 40n + 1 - k > 0$. There is therefore a unique highest-degree summand, and

$$K_n := \deg A_n = \frac{1091n^2 + 81n + 2}{2}, \quad W_n := \deg U_n = K_n - M_n + \sum_{\ell \leq 15n} (1 - \nu_\ell) \varphi(\ell).$$

For fixed n , the positive representation below gives $H_n(x) = O(1)$ as $x \rightarrow \infty$. Hence $\Lambda_n(x) = O(x^{W_n - K_n})$. Since $F(x) = O(x^{-1})$ and $K_n \geq 1$, the identity $V_n = U_n F - \Lambda_n$ gives $\deg V_n \leq W_n - 1$. Thus W_n clears both coordinates.

The limiting degree cost. The limits below are the cyclotomic limits of [4, Lemmas 1–2, p. 155]. The proof uses the argument through reciprocal intervals, the summatory totient estimate and the trigamma function from the proof of [3, Lemma 1, p. 466], with an explicit truncation of the block sum. The elementary summatory estimate $\sum_{\ell \leq y} \varphi(\ell) = 3y^2/\pi^2 + O(y \log y)$ gives

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \varphi(\ell) \rightarrow \frac{675}{\pi^2}.$$

For $[u, v] \in \mathcal{I}$, the condition $\{n/\ell\} \in [u, v]$ is the disjoint union of intervals $n/(k+v) < \ell \leq n/(k+u)$ for $k \geq 0$. For finitely many k the same summatory estimate applies term by term. The remaining terms have total normalised mass $O(1/K^2)$ after truncation at $k = K$, since they involve only $\ell \leq n/(K+u)$. Letting first n and then K tend to infinity proves

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \nu_\ell \varphi(\ell) \rightarrow \frac{3}{\pi^2} \sum_{[u,v] \in \mathcal{I}} \sum_{k \geq 0} \left(\frac{1}{(k+u)^2} - \frac{1}{(k+v)^2} \right) = \frac{3J}{\pi^2}.$$

Here no contributing index exceeds $14n$, since $u \geq 1/14$. Consequently

$$K_n/n^2 \rightarrow C_1, \quad (K_n - W_n)/n^2 \rightarrow C_0.$$

A positive remainder at every fixed real base. For $x > 1$ and $q = 1/x$, the source identity (9) on p. 156 has the positive representation

$$H_n(x) = \sum_{t \geq 0} q^{a_0 t} \frac{(q^{t+1}; q)_{a_1-1}}{(q; q)_{a_1-1}} \frac{(q; q)_{\beta-a_2-1}}{(q^{a_2+t}; q)_{\beta-a_2}}.$$

The last denominator has length $\beta - a_2$, the length given by the gamma expression (7) and the residues (8) of [4, p. 156]. The unnumbered display of $R(T)$ on that page prints the length $\beta - a_2 - 1$; the numbered identities fix the normalisation used here. Every finite product lies between $P = (q; q)_\infty > 0$ and 1, so

$$P^2 \leq H_n(x) \leq \frac{P^{-2}}{1 - q^{a_0}}.$$

In particular $\log H_n(x) = O_x(1)$ and $\Lambda_n(x) > 0$.

For fixed $x > 1$, the cyclotomic identity

$$\log \Phi_\ell(x) - \varphi(\ell) \log x = \sum_{d|\ell} \mu(d) \log(1 - x^{-\ell/d})$$

has total absolute error $O_x(n)$ over $\ell \leq 15n$. Indeed, it is at most $15nB(x)$, where

$$B(x) = \sum_{d \geq 1} \frac{-\log(1 - x^{-d})}{d} < \infty.$$

Since $0 \leq 1 - \nu_\ell \leq 1$, it follows that

$$\log \Lambda_n(x) = -(K_n - W_n) \log x + O_x(n).$$

Homogenisation. Because U_n, V_n are integral polynomials of degree at most W_n , $b^{W_n}U_n(a/b)$ and $b^{W_n}V_n(a/b)$ are integers. That integrality and the cleared linear-form identity it produces are [cleared linear form identity](#). Moreover

$$\begin{aligned} \log(b^{W_n}\Lambda_n(a/b)) &= K_n \log b - (K_n - W_n) \log a + O_{a/b}(n) \\ &= (C_1 \log b - C_0 \log a)n^2 + o(n^2). \end{aligned}$$

The coefficient is negative under the theorem's hypothesis. Thus positive integral linear forms in $F(a/b)$ tend to zero. If $F(a/b) = r/s$ were rational, every such form would have absolute value at least $1/|s|$, a contradiction. The separation bound for a nonzero integral form at a rational target is [rational integer linear form gap](#). $\square$

Corollary 2.2. $F((31/4)^r)$ is irrational for every integer $r \geq 1$.

Proof. The exact inequalities $31^2 < 4^5$ and $4^{200} < 31^{81}$ give $2/5 < \log 4 / \log 31 < 81/200$. The first term of each trigamma difference yields

$$J \geq J_0 := \sum_{(u,v) \in \mathcal{I}} (u^{-2} - v^{-2}) = \frac{2015640690251}{25971865920}.$$

Using $\pi > 157/50$ gives the rational lower bound

$$\theta^* > \frac{2359630009523263}{5820307922172744} > \frac{81}{200}.$$

Finally, the logarithmic ratio and coprimality are preserved by a common positive integer power. $\square$

Lean checks Theorem 2.1 as [the rational-base region](#), and Corollary 2.2 as [the powers of 31/4](#), with [the base 31/4](#) as the case $r = 1$. Both declarations apply the source supply rather than assuming it. The construction of the approximations is Zudilin's [4, 5]. What is checked here is the Lean formalisation of that construction's consequence.

Corollary 2.3 (an irrationality measure uniform over powers). *For coprime $a > b \geq 1$ with $\theta = \log b / \log a < \theta^*$ and every integer $r \geq 1$,*

$$\mu_{\text{irr}}(F((a/b)^r)) \leq \frac{1 - \theta}{\theta^* - \theta}.$$

Here $\mu_{\text{irr}}(\xi)$ is the supremum of the exponents ν for which $|\xi - p/q| < q^{-\nu}$ has infinitely many reduced rational solutions. In particular, $\mu_{\text{irr}}(F((31/4)^r)) < 301$ for every $r \geq 1$.

Proof. The passage from the forms to an exponent bound is the standard one that Zudilin uses at integer bases [4, p. 162]; the estimates below make it uniform in r . The raw source coefficient is a sum of $O(n)$ Laurent monomials times two Gaussian polynomials, each of coefficient sum at most 2^{27n+2} . Its coefficient norm is $\exp(O(n))$, and every exponent is at most K_n . The same normalising multiplier used above therefore gives

$$|U_n(x)| \leq x^{W_n} \exp(O_x(n)) \quad (x > 1 \text{ fixed}).$$

Set $\xi = F(a/b)$, $Q_n = b^{W_n} U_n(a/b)$ and $P_n = b^{W_n} V_n(a/b)$. With

$$\alpha = (C_1 - C_0) \log a, \quad \tau = C_0 \log a - C_1 \log b > 0,$$

we have $\log(Q_n \xi - P_n) = -\tau n^2 + o(n^2)$ and $|Q_n| \leq \exp(\alpha n^2 + o(n^2))$.

For integers A, B, p, q with $q > 0$, if $L = A\xi - B$ and $2q|L| \leq 1$, then

$$|L| \leq |A| |\xi - p/q|.$$

When $Ap - Bq = 0$ this is equality. Otherwise the nonzero integer $Ap - Bq$ gives $1/q \leq |L| + |A| |\xi - p/q|$, which proves the inequality. For a fixed $\eta \in (0, \tau)$ choose $n = \lceil \sqrt{\log(2q)/(\tau - \eta)} \rceil$ and apply it to (Q_n, P_n) . The two-sided remainder estimate yields

$$|\xi - p/q| \geq q^{-(\alpha + \tau + 2\eta)/(\tau - \eta) - o(1)}.$$

Let $\eta \downarrow 0$. The resulting bound is $1 + \alpha/\tau = (1 - \theta)/(\theta^* - \theta)$. Taking a common power multiplies α, τ by r , leaving this quotient unchanged; the constants in the approximation inequality may depend on r . For $31/4$, sixteen terms of each trigamma difference and $\pi > 314159/100000$ give $\theta^* > 0.40568$. The logarithm series gives $\log 4 / \log 31 < 0.4036982$. Thus the bound is less than $2981509/9909 < 301$. $\square$

Comparison and scope. Bundschuh and Väänänen's Theorem 2 at $\alpha = -1$ [2, p. 177] gives irrationality for $\log b / \log a < \theta_{BV} := 1/2 - 1/\pi^2$. Since $\pi^2 < 10$, one has $\theta_{BV} < 2/5 < \log 4 / \log 31$, so $31/4$ lies outside that sufficient region. The two sufficient regions differ on $[\theta_{BV}, \theta^*)$. For the generalized q -logarithm, Zudilin remarks that his results extend to non-integer rational bases $p = r/s$ under an assumption $\log |r| > c \log |s|$ for a computable constant $c > 0$, without computing c [5, Sec. 2, p. 4]. For F , the specialisation above shows that $c = \mu$, the exponent bound of [4, p. 162], is admissible.

Theorem 2.4 (the $7/2$ height condition). *The integer power certificate $2^{18} < 7^7$ yields the Archimedean height condition*

$$\frac{\log 7}{\log(7/2)} < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

This is the elementary parameter check at $q = 7/2$ in Bundschuh and Väänänen's Theorem 2 ($\alpha = -1$). Their analytic irrationality theorem is not a Lean result here.

Negative bases are not treated.

2.1 The base-uniform degree restriction

The preceding construction uses degree and decay estimates valid at every fixed real base greater than one. Such uniformity itself imposes a limit.

Theorem 2.5 (Archimedean cap on base-uniform rank-two families). *Let (U_n, V_n) be pairs in $\mathbb{Z}[X]^2$ satisfying $\Lambda_n(x) = U_n(x)F(x) - V_n(x) \neq 0$, $\deg U_n, \deg V_n \leq \delta n^2(1+o(1))$, $\log \max(H(U_n), H(V_n)) \leq hn^2(1+o(1))$ with H the ℓ^1 coefficient norm, and $\log |\Lambda_n(x)| = -\sigma n^2 \log x (1+o(1))$ for every real $x > 1$, with $\sigma, \delta > 0$ and $h \geq 0$ independent of x . Then with $d_n = \max(\deg U_n, \deg V_n)$, the homogenised forms $b^{d_n} \Lambda_n(a/b)$ tend to zero whenever $\log b / \log a < \sigma / (\sigma + \delta)$, and $\sigma / (\sigma + \delta) \leq 1/2$.*

Proof. Write $H_n = \max(H(U_n), H(V_n))$. Then $|U_n(x)|, |V_n(x)| \leq H_n x^{d_n}$ for real $x > 1$. Suppose $\sigma > \delta$ and choose an integer $p \geq 2$ with $(\sigma - \delta) \log p > h$. Set $a_n = U_n(p)$, $b_n = V_n(p)$ and $L_n = a_n F(p) - b_n$. Hypotheses on height and degree give $|a_n| \leq \exp((h + \delta \log p)n^2 + o(n^2))$, while $|L_n| = \exp(-\sigma \log p n^2 + o(n^2))$. The adjacent integer

$$a_n b_{n+1} - a_{n+1} b_n = a_{n+1} L_n - a_n L_{n+1}$$

is then $o(1)$, hence eventually zero. Also $a_n \neq 0$ for large n : otherwise the nonzero integer $L_n = -b_n$ would have absolute value less than 1. Thus b_n/a_n is eventually a fixed rational r . If $F(p) \neq r$ then $|L_n| \geq |F(p) - r|$; if $F(p) = r$ then $L_n = 0$. Both contradict the hypotheses, so $\sigma \leq \delta$. The integer argument of this paragraph, from the two cross-product limits to the contradiction, is [no small forms of cross product limits](#). The homogenised logarithm satisfies

$$\limsup_{n \rightarrow \infty} n^{-2} \log |b^{d_n} \Lambda_n(a/b)| \leq \delta \log b - \sigma \log(a/b),$$

which is negative on the stated sufficient region. The arithmetic form of that region is [logarithmic region iff negative balance](#), and the final numerical clause is [half cap iff](#). The conclusion bounds the sufficient cutoff furnished by the displayed degree estimate. An exclusion for a particular family requires its actual degree and remainder asymptotics. $\square$

3 Exact normalized-Hankel order in Zudilin's construction

Here the problem is cancellation inside a determinant. Entrywise orders give only a lower bound because terms of that order may cancel. The moment expansion below resolves this by finding one uniquely least-order index tuple, whose coefficient is nonzero. This is a formal-power-series argument; fixed-base estimates are a separate question addressed after the proof.

At $x = z = 1$, Zudilin's normalized moments [5, (6), pp. 6–7] are

$$v_m^* = \sum_{t \geq 0} q^{(m+1)t} \frac{(q; q)_m^3 (q^{t+1}; q)_m}{(q^{m+1+t}; q)_{m+1}}, \quad V_N^* = \det_{0 \leq i, j < N} (v_{i+j}^*).$$

His row transformation proves

$$\text{ord}_q V_N^* \geq \frac{N(N-1)(2N-1)}{6}$$

for every $N \geq 1$ [5, Sec. 4, Lemma 1, pp. 6–7]. A formal moment expansion identifies the unique least-order term and shows that this estimate is always sharp.

Theorem 3.1 (sharp normalized Hankel order). *For every $N \geq 1$,*

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6},$$

and the coefficient of the first nonzero monomial is

$$[q^{N(N-1)(2N-1)/6}]V_N^* = \frac{(N!)^2(N+1)!}{2^N}.$$

Proof. Write $P = (q; q)_\infty$ and introduce the coefficientwise formal series

$$G_q(w) = \frac{1}{(w; q)_\infty^3} \sum_{t \geq 0} \frac{w^t}{(q; q)_t} \frac{(q^t w^2; q)_\infty}{(q^t w; q)_\infty^2}, \quad a_k(q) = [w^k]P^4 G_q(w).$$

For fixed w -degree, all operations define power series in q . The product identities $(q; q)_m = P/(q^{m+1}; q)_\infty$ and $(q^a; q)_m = (q^a; q)_\infty / (q^{a+m}; q)_\infty$ give the exact formal identity

$$v_m^* = P^4 G_q(q^{m+1}) = \sum_{k \geq 0} a_k(q) q^{(m+1)k}.$$

At $q = 0$, the term $t = 0$ in G_q is $(1 - w^2)/(1 - w)^5$ and the terms $t \geq 1$ sum to $w/(1 - w)^4$. Consequently

$$G_0(w) = \frac{1 + 2w}{(1 - w)^4}, \quad a_k(0) = \frac{(k+1)^2(k+2)}{2} =: c_k > 0.$$

The expansion below is Heine's formula for a Hankel determinant of moments as a sum of squared Vandermonde products; Zudilin uses its integral form and reproduces its proof in [7, Sec. 2, (2)–(5), pp. 2–3]. Apply Cauchy–Binet to a finite truncation of the moment sum and then pass coefficientwise to the limit. This is legitimate because only finitely many increasing index tuples contribute to any fixed q -degree. It gives

$$V_N^* = \sum_{k_0 < \dots < k_{N-1}} \left(\prod_{i=0}^{N-1} a_{k_i}(q) q^{k_i} \right) \prod_{0 \leq i < j < N} (q^{k_i} - q^{k_j})^2.$$

The summand indexed by (k_i) has order

$$\sum_{i=0}^{N-1} (2N - 1 - 2i)k_i.$$

Each $a_k(q)$ has q -order zero with positive initial coefficient c_k , and $k_i \geq i$. Equality with the least possible order occurs uniquely when $k_i = i$ for every i . That tuple contributes

$$\sum_{i=0}^{N-1} (2N - 1 - 2i)i = \frac{N(N-1)(2N-1)}{6}$$

and leading coefficient

$$\prod_{i=0}^{N-1} c_i = \frac{(N!)^2 (N+1)!}{2^N}.$$

No other tuple can cancel this coefficient. $\square$

Lean checks both identities at every rank as [the exact order](#) and [the leading coefficient](#).

Formal order and fixed-base size. The theorem identifies the first formal term. Formal order alone would not control a fixed- q residual: multiplying by $(1-q)^{N^3}$ preserves that first term and changes the logarithm by a cubic quantity at fixed $0 < q < 1$. The separate positive-measure argument for these same moments proves $V_N^*(q) > 0$ and $\log(V_N^*(q)/(C_N q^{B_N})) = O_q(N)$, where $B_N = N(N-1)(2N-1)/6$ and $C_N = (N!)^2 (N+1)!/2^N$. Its complete ordinary proof is in `ZudilinHankelPositiveMeasure.md` and the long record. Neither fact supplies a denominator factor for the 2004 polynomial forms.

4 Local cancellation and the remaining real estimate

The tools of this section are elementary: congruences at the endpoints, the pigeonhole principle and Bézout's identity for unimodular rows. The hypotheses on minors, multiplicities and real remainders are stated explicitly where they enter.

Theorem 4.1 (no corridor at base $3/2$). *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D , the tuple $(3, 2, N, K, Q, D)$ is not a [coordinatewise corridor](#).*

Theorem 4.2 (cleared-tail recurrence). *Let $r, s, B, F \in \mathbb{Q}$ with $r \neq 0$, let $c : \mathbb{N} \rightarrow \mathbb{Q}$, and let P_N and U_N be the cleared tail state. Then for every N the [cleared-tail recurrence](#) is*

$$U_{N+1} = r U_N - B c(N+1) s^{N+1}.$$

Theorem 4.3 (the forcing term). *Let s, B be natural numbers and $c : \mathbb{N} \rightarrow \mathbb{N}$, and put $G_N = B c(N+1) s^{N+1}$.*

1. *If $s \geq 2, B \geq 1$ and $c(N+1) \geq 1$, then $2^{N+1} \leq G_N$.*
2. *If $s = 1$, then $G_N = B c(N+1)$.*

Fix a common width W . For $P \in \mathbb{Z}[X]$ of degree at most W , write

$$H_W(P) = 2^W P(3/2) = \sum_{i=0}^W [P]_i 3^i 2^{W-i}.$$

A specialised row is primitive when its two integer coordinates have gcd one. This normalisation is different from removing the common polynomial coefficient content, and must precede the local count. For depths $R, S \geq 0$, define

$$J_{3,R}(P) = H_W(P) \pmod{3^R}, \quad J_{2,S}(P) = H_W(P) \pmod{2^S}.$$

Lemma 4.4 (bottom-jet divisibility). *Vanishing of the [bottom jet](#) is exactly divisibility by the corresponding power of three: $J_{3,R}(P) = 0$ if and only if $3^R \mid H_W(P)$.*

All four jets of (U, V) vanish precisely when $D = 3^R 2^S$ divides both specialised coordinates. The dependence on the declared width remains fixed when rows are added.

Theorem 4.5 (binary four-jet collision). *Fix a width W and depths R, S , and let $(U_j, V_j)_{j < M}$ be any M pairs of integral polynomials. Call a subset of $\{0, \dots, M-1\}$, equivalently a vector of $\{0, 1\}^M$, a binary selector. If the 2^M binary selectors outnumber the finite four-jet target*

$$(\mathbb{Z}/3^R \mathbb{Z})^2 \times (\mathbb{Z}/2^S \mathbb{Z})^2,$$

then two distinct subsets have the same four-jet sum. Subtracting their indicator vectors gives a nonzero coefficient vector in $\{-1, 0, 1\}^M$ cancelling all four jets. The target has exact cardinality

$$(3^R)^2 (2^S)^2.$$

In particular, if $R > 0$ and $4R + 2S \leq M$, such a collision exists.

Proof. Send each binary selector to the sum of the four-jet signatures it selects. The claimed cardinal inequality and the pigeonhole principle give two distinct selectors in the same fibre. The cardinality formula is the product of the four cyclic-modulus cardinalities. For $R > 0$,

$$(3^R)^2 (2^S)^2 < (4^R)^2 (2^S)^2 = 2^{4R+2S} \leq 2^M,$$

which proves the stated sufficient threshold. The cardinality formula, the collision, the signed $\{-1, 0, 1\}$ vector and the sufficient width are together [four jet paper statement](#). $\square$

The ambient count does not use relations between the two residue coordinates. Vanishing minors can reduce that cost to one coordinate.

Theorem 4.6 (Bézout–Plücker tail collapse). *Let R_0 be a commutative ring and let $w_n = (A_n, B_n) \in R_0^2$. Suppose that every row is unimodular ($u_n A_n + v_n B_n = 1$ for some u_n, v_n) and every adjacent minor vanishes:*

$$A_n B_{n+1} - B_n A_{n+1} = 0 \quad (n \geq 0).$$

Then every pairwise minor $A_i B_j - B_i A_j$ vanishes. In particular, take $R_0 = \mathbb{Z}/(2^S 3^R) \mathbb{Z}$ with $R > 0$. If $S + 2R \leq k$, there are two distinct binary selectors $s, t \in \{0, 1\}^k$ such that

$$\sum_{i < k} s_i w_i = \sum_{i < k} t_i w_i.$$

Thus the sufficient width is $S + 2R$, rather than the ambient two-coordinate width $2S + 4R$.

Proof. A Bézout identity makes each row a unimodular anchor; a unit coordinate is the special case already recorded. Vanishing of the next minor therefore writes the next row as a scalar multiple of the current one; induction places the entire tail on the line through w_0 and proves the pairwise-minor assertion. A determinant-one Bézout shear sends w_0 to $(1, 0)$, so all selector sums have only one free residue coordinate and occupy at most $2^S 3^R$ values. Finally

$$2^S 3^R < 2^S 4^R = 2^{S+2R} \leq 2^k,$$

and pigeonhole gives the two selectors. The ring-generic minor collapse and the modular selector collision are together [plucker paper statement](#). $\square$

Primitive integer rows are unimodular modulo every modulus. A particular coordinate need not be a unit: $(2, 3)$ modulo 6 is an example. The minor-vanishing hypothesis remains a separate condition on the source rows.

4.1 One minor gcd controls two different costs

The next proposition compares two explicit exponent polynomials of a Padé-type denominator model. No coefficient formula producing them is derived here, so the proposition by itself gives no integrality, nonvanishing or approximation estimate.

Proposition 4.7 (exponent model: summand bound and exact gap). *Let $\tilde{E}_n = 3n^2 - n$ and put*

$$\begin{aligned}\tilde{P}(n, k) &= 2(k(n - k) + nk) + k(k - 1), \\ \tilde{Q}(n, m) &= 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \quad j = n - m - 1.\end{aligned}$$

Then, for integers n, k, m :

1. *if $0 \leq k \leq n$, the [summand exponent bound](#) is $\tilde{P}(n, k) \leq \tilde{E}_n$, and the gap factors as $\tilde{E}_n - \tilde{P}(n, k) = (n - k)(3n - k - 1)$;*
2. *the [exact gap identity](#) is $\tilde{E}_n - \tilde{Q}(n, m) = 2(n + m(m - 1))$.*

Proposition 4.8 (row-content determinant scaling). *Rowwise integer contents scale the [exterior determinant](#) by the same factors: the [content factorisation](#) and the [absolute determinant scaling](#) identify the local divisor with the Archimedean height cost.*

Let a rank-two lattice $\Lambda \subset \mathbb{Z}^2$ be generated by primitive rows, and let $g > 0$ be the gcd of their 2×2 minors. By the description of Smith invariants through gcds of minors [6, Thms. 2.3–2.4, p. 3], its Smith invariants are 1, g . Therefore

$$|\text{im}(\Lambda \bmod D)| = \frac{D^2}{\gcd(g, D)}, \quad \left[\mathbb{Z}^2 : \frac{\Lambda \cap D\mathbb{Z}^2}{D} \right] = \frac{g}{\gcd(g, D)}.$$

To verify the formulas, make a unimodular change of coordinates sending Λ to $\mathbb{Z} \oplus g\mathbb{Z}$; that change preserves $D\mathbb{Z}^2$. The first quotient counts the modular signatures. The second is the index remaining after dividing a successful collision by D . Once $D \mid g$, the modular count is D , while the remaining index is g/D .

If two independent divided rows (A_i, B_i) have $|A_i| \leq H$ and $|A_i\tilde{\zeta} - B_i| \leq \varepsilon$, their determinant gives

$$2H\varepsilon \geq \frac{g}{\gcd(g, D)}.$$

This is a restriction on two independent forms in the divided lattice. For one nonzero integral form tending to zero, no additional product $H\varepsilon \rightarrow 0$ is required.

4.2 A collision must have a small nonzero real remainder

A real bin records the analytic requirement alongside the modular signature. The multiplicity to control is the number of equal real values within one modular fibre.

Theorem 4.9 (quantitative bounded-fibre escape). *Let A, B, J be finite sets and let $f : A \rightarrow B$, $g : A \rightarrow \mathbb{R}$ and $\iota : A \rightarrow J$. Suppose that each simultaneous fibre of (f, g) has at most k elements and that, for some $\delta > 0$,*

$$\iota(x) = \iota(y) \implies |g(x) - g(y)| < \delta.$$

If $|B||J|k < |A|$, then some distinct $x, y \in A$ satisfy

$$f(x) = f(y), \quad 0 < |g(x) - g(y)| < \delta.$$

Proof. Partition A by (f, ι) . Some cell has more than k elements, so its g -values cannot all agree. Two unequal values in that cell have the same modular signature and differ by less than δ . $\square$

For primitive integer rows (A_j, B_j) , put $e_j = A_j F(3/2) - B_j$ and apply the theorem to binary selectors, with

$$f(\varepsilon) = \sum_j \varepsilon_j (A_j, B_j) \pmod{D}, \quad g(\varepsilon) = \sum_j \varepsilon_j e_j.$$

Let Q be the number of attained modular signatures and let k bound the simultaneous (f, g) fibres. All selector remainders lie in an interval of length $T = \sum_j |e_j|$. Dividing that interval into half-open bins of width D/n gives the sufficient inequality

$$2^M > Qk \left(\left\lfloor \frac{nT}{D} \right\rfloor + 1 \right). \quad (1)$$

Its conclusion is a signed row sum divisible coordinatewise by D , with a nonzero divided real remainder of absolute value less than $1/n$. More precisely, if modular fibre b has its own real span T_b and exact value multiplicity k_b , it suffices that

$$2^M > \sum_b k_b \left(\left\lfloor \frac{nT_b}{D} \right\rfloor + 1 \right).$$

These are estimates for the primitive real remainders of the declared source family. Bounded unnormalised hypergeometric remainders cannot replace them. A nonzero function need not be nonzero at $3/2$, as the factor $2X - 3$ shows; positive individual remainders need not remain positive after subtraction.

4.3 The source-specific endpoint question

The following sufficient construction keeps source membership, primitive scaling, local cancellation and the real estimate together. Without source and height restrictions, arbitrary polynomial lifts of small rational approximations would simply restate irrationality.

Problem 4.10 (common-width simultaneous endpoint-jet construction). *Exhibit an integer constant $C \geq 1$ and, for every sufficiently large positive integer n , positive integers W_n, R_n, S_n, M_n such that*

$$n^2 \leq W_n, R_n, S_n \leq Cn^2, \quad 4R_n + 2S_n \leq M_n \leq Cn^2,$$

together with polynomial pairs coming from a named source family with an explicit coefficient-height bound after primitive normalisation, $(U_{n,j}, V_{n,j}) \in \mathbb{Z}[X]^2$ for $0 \leq j < M_n$, each of degree at most the common declared width W_n , whose specialised integer rows are primitive:

$$\gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j})) = 1.$$

Find a nonzero vector $\lambda^{(n)} \in \{-1, 0, 1\}^{M_n}$ for which, on putting

$$U_n = \sum_{j < M_n} \lambda_j^{(n)} U_{n,j}, \quad V_n = \sum_{j < M_n} \lambda_j^{(n)} V_{n,j},$$

the pair (U_n, V_n) is not $(0, 0)$, all four common-width jets vanish,

$$J_{3,R_n}(U_n) = J_{3,R_n}(V_n) = 0, \quad J_{2,S_n}(U_n) = J_{2,S_n}(V_n) = 0,$$

where every jet in this display is formed using the declared width W_n , and the resulting divided integer linear form

$$A_n = \frac{H_{W_n}(U_n)}{3^{R_n} 2^{S_n}}, \quad B_n = \frac{H_{W_n}(V_n)}{3^{R_n} 2^{S_n}}, \quad \rho_n = A_n F(3/2) - B_n$$

satisfies the explicit analytic condition

$$0 < |\rho_n| < \frac{1}{n}.$$

The jet equations make A_n, B_n integers. If $F(3/2) = a/b$ were rational, a nonzero ρ_n would have absolute value at least $1/|b|$, contradicting $|\rho_n| < 1/n$ for large n . The required analytic inequality is exactly the displayed scalar condition. For irrationality it suffices to obtain such forms along any unbounded sequence of n ; the all-sufficiently-large- n formulation is a stronger construction requirement. Height estimates become relevant when a specific construction uses them to obtain this scalar inequality, or when an independent-row determinant is invoked.

For the literal two-parameter source deformations, the long record keeps the row-wise primitive normalisation, the two local minor valuations and the remaining real-error minimum together. The narrow regular-scale family has an ordinary determinant obstruction to bounded divided errors; that obstruction does not exclude sparse scales or the wider deformation. Thus the next estimate must concern the real remainders of the chosen family. Increasing the number of modular collisions alone does not establish it.

Functional equations. Bell and Smertnig's classification of Mahler series with multiplicative coefficients shows that $L(z) = \sum_{n \geq 1} \tau(n) z^n$ is not k -Mahler for any $k \geq 2$ [9, Thm. 1.3 and the consequences on p. 3]. The earlier simultaneous-2/3 obstruction is consequently subsumed by this known single-base result. A construction using additional functions or functional relations must specify those functions and its closure conditions; the single-base statement is not an obstruction to every approximation method.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. The ordinary proofs used here are printed with their hypotheses.

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A Guide to the formal sources

The seven modules named below are linked at the pinned source revision 99f4bf47422a. Additional declarations in the body of the note are linked at a second immutable revision, f36a98bf3d3e6f65f1074e3b800e3293d5b8a51a. These are separate snapshots, not one checked revision of the whole library. Each hyperlink opens the named declaration at the commit recorded in its URL.

The declarations linked at 99f4bf47422a live in seven modules: RationalBaseLambert, QApéryDiagonalNonEquivalence, RationalPadéArithmetic, ZudilinConeArithmetic, ZudilinHeightRegion, HermitePadéNoGo, and BezoutPluckerJets. The first contains the corridor, cleared-tail recurrence, and elementary $7/2$ certificate; the second checks the finite $n = 0$ diagonal residual; the remaining four separate the Padé exponent arithmetic, endpoint arithmetic, logarithmic comparisons, rectangular exponent model, and Bézout–Plücker tail collapse. The link coordinates for those seven modules are validated against that pinned revision, so they remain correct as later work moves lines in the working tree.

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Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.